

Machine Code Generation

- ⇒ Code generation is the last phase of compiler construction.
- ⇒ input: Intermediate code (optimized code).
- ⇒ output: Target machine code.

Target language:

- ⇒ For designing the good code generator it is necessary to have prior knowledge of target machine and instruction set used for this target machine.
- ⇒ Generally, the assembly language is the target code.
- ⇒ The target computer addresses are given in bytes and four bytes from a word.
- ⇒ There are n general purpose registers $R_0, R_1, \dots, R_{n-1}$.
- ⇒ The two address instruction is of the form:

op source destination

- The op is an opcode.
- The source and destination are specified by registers and memory locations.
- The addressing modes used are as follows:

Addressing mode	Form	Address	Added cost
absolute	M	M	1
register	R	R	0
indexed	c(R)	c + contents(R)	1
indirect register	*R	contents(R)	0
indexed indexed	*c(R)	contents(c + contents(R))	1
literal	#C	C	1

Cost of the instruction:

- ⇒ The instruction cost can be computed as one plus and cost associated with the source and destination addressing modes given by added cost.

Date: _____

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Instruction	Cost	Interpretation
MOV R ₀ , R ₁	1	Cost of register mode + 1 = 0 + 1 = 1
MOV R ₁ , M	2	use of memory variable + 1 = 1 + 1 = 2
SUB S(R ₀), *O(R ₁)	3	use of first constant + use of second constant + 1 = 2

Example:

Compute the cost of following set of instructions.

MOV a, R₀ADD b, R₀MOV R₀, CMOV a, R₀ 2ADD b, R₀ 2MOV R₀, C 2

Total cost = 6

⇒ The cost of this set is 6

Example: Compute the cost of following set of instructions.

MOV *R₁, *R₀ADD *R₂, *R₀MOV *R₁, *R₀ 1ADD *R₂, *R₀ 1

Total cost = 2

① Compute the cost.

MOV B, R₀ADD C, R₀MOV R₀, A

For instruction:

MOV B, R₀

↓ ↓ ↘
1 1 0
Cost of instruction cost of memory location cost of register

MOV B, R₀ → Cost = 1 + 1 + 0 = 2ADD C, R₀ → Cost = 1 + 1 + 0 = 2MOV R₀, A → Cost = 1 + 0 + 1 = 2

Q Convert into machine instruction:

$$x = y - z$$

LD R_1, y

SUB R_1, R_1, z ($R_1 = R_1 - z$)

ST x, R_1 ($x = R_1$)

Q. Get an array value to variable.

$$b = a[i]$$

Suppose an array whose elements are 8-bytes

0	7	8	15	16	23	24	31	32	39	40	47
7	8	6	2	0	3						
0	1	2	3	4	5						

LD R_1, i ($R_1 = i$)

MUL $R_1, R_1, 8$ ($R_1 = R_1 * 8$)

LD $R_2, a(R_1)$ ($R_2 = a[R_1/8]$)

ST b, R_2 ($b = R_2$)